Q2

(a) <u>First part of the FTC.</u>

If f is a continuous integrable function on $\mathbb{R}$ and F is defined as

$$F(x) = \int_0^x f(t)\,dt$$

then $F(x)$ is differentiable and

$$F'(x) = f(x).$$

<u>Second part of the FTC.</u>

If f is continuous and integrable on $\mathbb{R}$ and F is an anti-derivative for f.

i.e. $F'(x) = f(x)$, then

$$\int_a^b f(x)\,dx = F(b) - F(a)$$

Applying the first part to the given problem. The function $f(t) = t^2 + \cos(t)$ is continuous and integrable on $\mathbb{R}$ so by the first part of the FTC. If

$$F(x) = \int_0^x t^2 + \cos(t)\,dt.$$

then $F'(x) = f(x) = x^2 + \cos(x)$

so $F'(\pi/2) = \left(\frac{\pi}{2}\right)^2 + \cos(\pi/2)$

$$= \frac{\pi^2}{4}.$$

(b) A sequence $\{x_n\}_{n=1}^{\infty}$ is convergent with limit $L \in \mathbb{R}$ if and only if for all $\varepsilon > 0$ there exists $N \in \mathbb{N}$ such that $|x_n - L| < \varepsilon$ whenever $n \geq N$.

OR in symbolic form

$$\forall \varepsilon > 0 \ \exists N \in \mathbb{N} \quad n \geq N \Rightarrow |x_n - L| < \varepsilon.$$

(c) Let's assume that $\alpha, \beta \neq 0$.

Let $\varepsilon > 0$ be given and by the convergence of $a_n \to a$ and $b_n \to b$ we can choose $N_1, N_2 \in \mathbb{N}$ so that

- $n \geq N_1 \Rightarrow |a_n - a| < \frac{\varepsilon}{2|\alpha|}$
- $n \geq N_2 \Rightarrow |b_n - b| < \frac{\varepsilon}{2|\beta|}$

$\boxed{\text{Let } N = \max(N_1, N_2)}$. So now

whenever $\boxed{n \geq N}$

$$\underbrace{|(\alpha a_n + \beta b_n) - (\alpha a + \beta b)|}_{\text{"}<\varepsilon\text{"}}$$

$$= |\alpha(a_n - a) + \beta(b_n - b)|$$

$$\leq |\alpha(a_n - a)| + |\beta(b_n - b)|,$$ by the Δ inequality for abs. value

$$= |\alpha| \underbrace{|a_n - a|} + |\beta| \underbrace{|b_n - b|}$$

since abs. val. is multiplicative, i.e. $|xy| = |x||y|$

$$< |\alpha| \frac{\varepsilon}{2|\alpha|} + |\beta| \frac{\varepsilon}{2|\beta|}$$

$$= \varepsilon$$

This shows $\{\alpha a_n + \beta b_n\}_{n=1}^{\infty}$ satisfies the definition for convergence to $\alpha a + \beta b$.

d) Consider an infinite series.

$\sum_{n=1}^{\infty} a_n$. It's k^{th} partial sum S_k

is the finite sum $\sum_{n=1}^{k} a_n = S_k$.

The formal definition of $\sum_{n=1}^{\infty} a_n$

$$\boxed{\sum_{n=1}^{\infty} a_n := \lim_{k \to \infty} (S_k)}.$$

$$\sum_{n=1}^{\infty} a_n = \lim_{k \to \infty} \left(\sum_{n=1}^{k} a_n \right)$$

(e) Consider the partial sum S_k.

$$S_k = \sum_{n=1}^{k} (\alpha a_n + \beta b_n)$$

$$= \left(\sum_{n=1}^{k} \alpha a_n \right) + \left(\sum_{n=1}^{k} \beta b_n \right)$$

, commutativity of addition.

$$= \alpha\left(\sum_{n=1}^{k} a_n\right) + \beta\left(\sum_{n=1}^{k} b_n\right)$$

Now we can take the limit as $k \to \infty$.

$$\sum_{n=1}^{\infty} (\alpha a_n + \beta b_n)$$

$$= \lim_{k\to\infty} S_k$$

$$= \lim_{k\to\infty} \left(\alpha\left(\sum_{n=1}^{k} a_n\right) + \beta\left(\sum_{n=1}^{k} b_n\right)\right)$$

$$= \alpha \lim_{k\to\infty}\left(\sum_{n=1}^{k} a_n\right) + \beta \lim_{k\to\infty}\left(\sum_{n=1}^{k} b_n\right)$$

, by part (c), linearity of sequence convergence

$$= \alpha a + \beta b,$$ substituting in the assumed infinite sums a, b.

(Continue at 11:58).

Q3 (a).

The ~~derivative~~ derivative $f'(a)$ of a function $f(x)$ is defined as

$$f'(a) = \lim_{h \to 0} \frac{f(a+h) - f(a)}{h}$$

, provided this limit exists.

When f is the function $f'(x) = x^6$.

Then

$$f'(a) = \lim_{h \to 0} \left(\frac{(a+h)^6 - a^6}{h} \right)$$

$$= \lim_{h \to 0} \left[\frac{\left(\cancel{a^6} + 6a^5 h + 15a^4 h^2 + 20a^3 h^3 + 15a^2 h^4 + 6ah^5 + h^6 - \cancel{a^6} \right)}{h} \right]$$

Pascal's Δ.

1
1 1
1 2 1
1 3 3 1
1 4 6 4 1
1 5 10 10 5 1
1 6 15 20 15 6 1

$$= \lim_{h \to 0} \left(6a^5 + 15a^4 h + 20a^3 h^2 + 15a^2 h^3 + 6ah^4 + h^5 \right).$$

$= 6a^5$, by the linearity of limits of functions as there is a factor of h in all the other terms

This holds for all $a \in \mathbb{R}$, $f'(a) = 6a^5$

So we can say the derivative is the function defined by

$$f'(x) = 6x^5.$$

b) $$g(x) = x^4 \sin(2x)$$

$$g'(x) = \frac{d}{dx}(x^4)\sin(2x) + x^4 \frac{d}{dx}(\sin(2x))$$

, by the product rule.

$$= 4x^3 \sin(2x) + x^4 \cos(2x).2$$

, using standard derivative of powers result and chain rule on $\sin(2x)$.

For the second derivative we apply these results again along with linearity to get

$$g''(x) = \frac{d}{dx}\left(4x^3 \sin(2x)\right) + \frac{d}{dx}\left(2x^4 \cos(2x)\right)$$

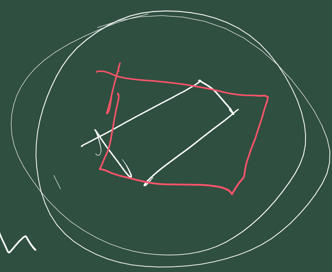
, by linearity.

$$= 12x^2 \sin(2x) + 4x^3 \cos(2x)\, 2 + 8x^3 \cos(2x) + 2x^4(-\sin(2x))\, 2.$$

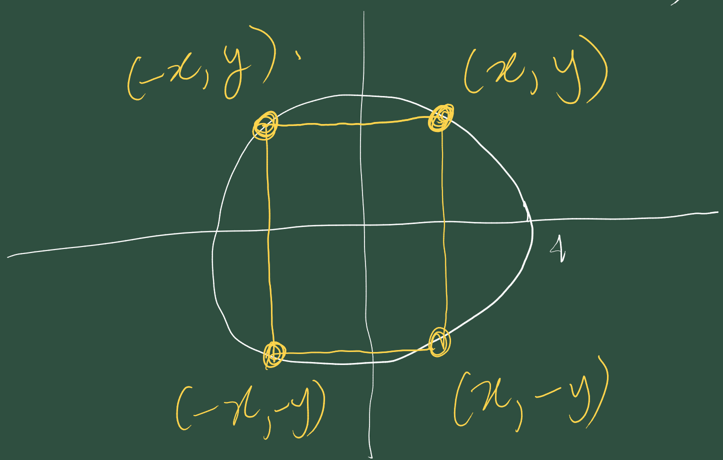
, using product and chain rules like before.

$$= (-4x^4 + 12x^2)\sin(2x) + 16x^3\cos(2x).$$

(c)



Centre the unit circle on the origin and rotate the rectangle so that it aligns with the axes. A rectangle of maximal area will touch the circle, at (x, y).



The area A of this rectangle is $A(x, y) = 2x\,2y = 4xy$.

(x,y) is constrained to lie on the unit circle, which has equation

$$1 = x^2 + y^2 \quad \text{and} \quad -1 \le x \le 1$$
$$\text{and } -1 \le y \le 1$$

$$\Rightarrow \quad y = \sqrt{1-x^2}$$

So
$$A(x,y) = 4xy$$
$$= 4x\sqrt{1-x^2}$$
$$= 4\sqrt{x^2-x^4}, \qquad -1 \le x \le 1$$

Which gives A as a single-variable function.

The maximal area is achieved at a critical point of A, i.e. a solution to

$$A'(x) = 0$$

$$\Leftrightarrow \cancel{4}\,\cancel{\tfrac{1}{2}}\left(x^2+x^4\right)^{-1/2}\left(\cancel{2}x-\overset{2}{\cancel{4}}x^3\right) = 0$$

$$\Leftrightarrow \frac{1-2x^2}{\left(x^2+x^4\right)^{1/2}} = 0.$$

$$\Leftrightarrow \quad 1-2x^2 = 0$$

$$\iff \quad x^2 = \frac{1}{2}$$

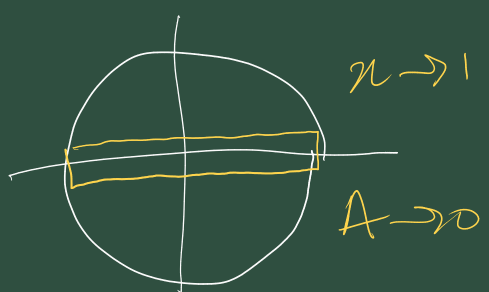
$$\iff \quad x = \pm\sqrt{1/2}.$$

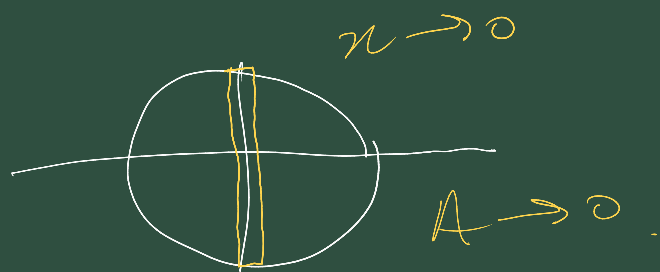
$$\iff \quad x = \frac{1}{\sqrt{2}},$$ as x is chosen to lie in the half-plane $x \geq 0$.

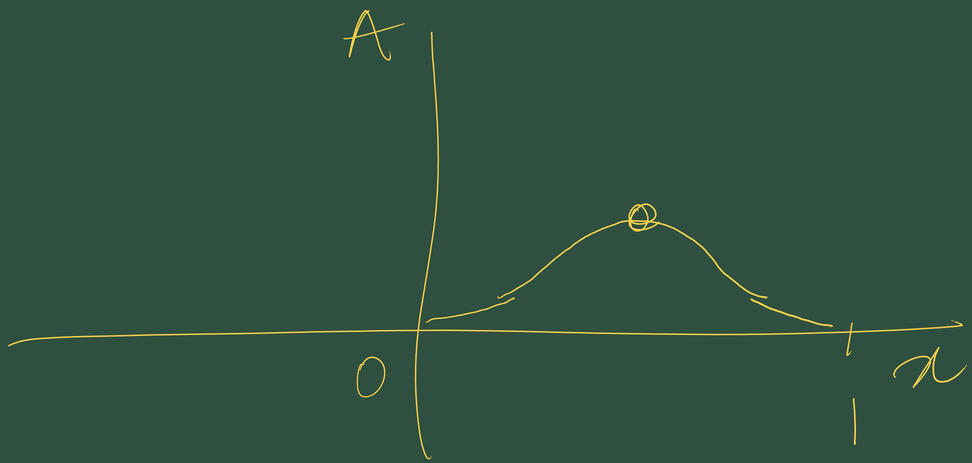
So we a single critical point $x = \frac{1}{\sqrt{2}}$ with corresponding y-value.

$$y = \sqrt{1 - x^2} = \frac{1}{\sqrt{2}}.$$

Clearly this is a maximum critical point as there are rectangles of ~~arbitrary~~ arbitrarily small formed by using x very close to 1, or 0.







So the maximal rectangle is actually the square of side length $2 \frac{1}{\sqrt{2}}$.

$= \sqrt{2}$,

(d). Find

$$\frac{d}{dx}\left(\frac{uv}{wz}\right)$$

$$= \frac{\frac{d}{dx}(uv)\,wz - uv\,\frac{d}{dx}(wz)}{(wz)^2}$$

, by the quotient rule.

$$= \frac{(u'v + uv')wz - uv(w'z + wz')}{(wz)^2}$$

, by the product rule.

$$= \frac{u'vwz + uv'wz - uvw'z - uvwz'}{(wz)^2}$$